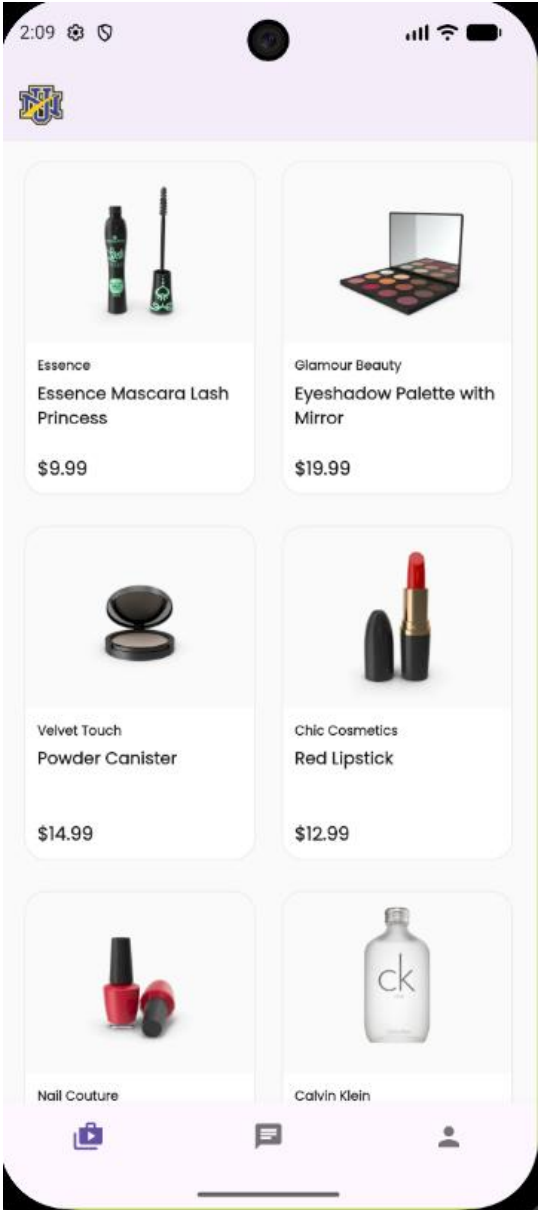


Lab Activity 2

Submission Form

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Place output based on the source code.



The screenshot shows a mobile application interface with a purple header containing the NU logo. Below the header is a grid of six product cards. Each card features an image of a beauty product, its name, and its price. The products are: Essence Mascara Lash Princess (\$9.99), Glamour Beauty Eyeshadow Palette with Mirror (\$19.99), Velvet Touch Powder Canister (\$14.99), Chic Cosmetics Red Lipstick (\$12.99), Nail Couture (price not visible), and Calvin Klein (price not visible). The bottom of the screen has a navigation bar with three icons: a shopping bag, a speech bubble, and a user profile.

Place your sample code and output based on the enhancement. You may add rows or cells if needed.



Source code –

product_screen.dart

```
import 'package:flutter/material.dart';
import 'package:flutter_screenutil/flutter_screenutil.dart';
import '../constants.dart';
import '../models/product.dart';
import '../services/product_service.dart';
import '../widgets/custom_text.dart';

class ProductScreen extends StatefulWidget {
  const ProductScreen({super.key});

  @override
  State<ProductScreen> createState() => _ProductScreenState();
}

class _ProductScreenState extends State<ProductScreen> {
  final ProductService _productService = ProductService();

  List<Product> _allProducts = [];
  bool _isLoading = true;
  String _errorMessage = '';
  String _searchQuery = '';

  @override
  void initState() {
    super.initState();
    _loadProducts();
  }

  Future<void> _loadProducts() async {
    setState(() {
      _isLoading = true;
      _errorMessage = '';
    });

    try {
      final products = await _productService.getAllProducts();
      if (!mounted) return;
      setState(() {
        _allProducts = products;
        _isLoading = false;
      });
    } catch (e) {
      if (!mounted) return;
      setState(() {
        _isLoading = false;
        _errorMessage = e.toString();
      });
    }
  }

  List<Product> get _filteredProducts {
    if (_searchQuery.isEmpty) {
      return _allProducts;
    }
    return _allProducts
      .where((p) =>
        p.title.toLowerCase().contains(_searchQuery.toLowerCase()) ||
        p.description.toLowerCase().contains(_searchQuery.toLowerCase()) ||
        p.brand.toLowerCase().contains(_searchQuery.toLowerCase())
      ).toList();
  }

  @override
  Widget build(BuildContext context) {
    final theme = Theme.of(context);
    final isDark = theme.brightness == Brightness.dark;

    return Scaffold(
      backgroundColor: theme.scaffoldBackgroundColor,
      body: SafeArea(
        child: Column(
          // Search Bar
          children: [
            Padding(
              padding: EdgeInsets.symmetric(horizontal: 16.w, vertical: 12.h),
              child: Container(
                height: 48.h,
                decoration: BoxDecoration(
                  color: theme.cardTheme.color,
                  borderRadius: BorderRadius.circular(12.r),
                  border: Border.all(
                    color: isDark ? AppColors.darkBorder : AppColors.lightBorder,
                  ),
                ),
                child: TextField(
                  onChanged: (val) {
                    setState(() {
                      _searchQuery = val;
                    });
                  },
                  decoration: InputDecoration(
                    hintText: 'Search products...',
                    hintStyle: TextStyle(
                      fontSize: 14.sp,
                      color: isDark ? AppColors.darkMuted : AppColors.lightMuted,
                      fontFamily: 'Poppins',
                    ),
                    prefixIcon: Icon(
                      Icons.search,
                      color: isDark ? AppColors.darkMuted : AppColors.lightMuted,
                      size: 28.sp,
                    ),
                    border: InputBorder.none,
                    contentPadding: EdgeInsets.symmetric(vertical: 12.h),
                  ),
                ),
              ),
            ),
            Expanded(
              child: _buildMainContent(theme, isDark),
            ),
          ],
        ),
      ),
    );
  }

  Widget _buildMainContent(ThemeData theme, bool isDark) {
    if (_isLoading) {
      return _buildSkeletonGrid(isDark);
    }

    if (_errorMessage.isNotEmpty) {
      return Center(
        child: Padding(
          padding: EdgeInsets.all(24.r),
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              Icon(
                Icons.wifi_off_rounded,
                size: 64.sp,
                color: isDark ? AppColors.darkMuted : AppColors.lightMuted,
              ),
              SizedBox(height: 16.h),
              CustomText(
                text: 'Oops! Something went wrong',
                fontSize: 16.sp,
                fontWeight: FontWeight.bold,
              ),
              SizedBox(height: 8.h),
              CustomText(
                text: _errorMessage,
                fontSize: 13.sp,
                textAlign: TextAlign.center,
              ),
              SizedBox(height: 28.h),
              ElevatedButton.icon(
                onPressed: _loadProducts,
                icon: const Icon(Icons.refresh),
                label: const Text('Try Again'),
                style: ElevatedButton.styleFrom(
                  shape: RoundedRectangleBorder(
                    borderRadius: BorderRadius.circular(8.r),
                  ),
                ),
              ),
            ],
          ),
        ),
      );
    }
  }
}
```



```
final filtered = _filteredProducts;

if (filtered.isEmpty) {
  return Center(
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        Icon(
          Icons.search_off_rounded,
          size: 64.sp,
          color: isDark ? AppColors.darkMuted : AppColors.lightMuted,
        ),
        SizedBox(height: 16.h),
        CustomText(
          text: 'No products found',
          fontSize: 16.sp,
          fontWeight: FontWeight.bold,
        ),
      ],
    ),
  );
}

return RefreshIndicator(
  onRefresh: _loadProducts,
  child: GridView.builder(
    padding: EdgeInsets.symmetric(horizontal: 16.w, vertical: 8.h),
    gridDelegate: SliverGridDelegateWithFixedCrossAxisCount(
      crossAxisCount: 2,
      childAspectRatio: 0.65,
      crossAxisSpacing: 12.w,
      mainAxisSpacing: 12.h,
    ),
    itemCount: filtered.length,
    itemBuilder: (context, index) {
      final product = filtered[index];
      return _buildProductCard(context, product, theme, isDark);
    },
  ),
);

Widget _buildProductCard(BuildContext context, Product product, ThemeData theme, bool isDark) {
  return Card(
    elevation: 0,
    color: theme.cardTheme.color,
    shape: RoundedRectangleBorder(
      borderRadius: BorderRadius.circular(16.r),
      side: BorderSide(
        color: isDark ? AppColors.darkBorder : AppColors.lightBorder,
      ),
    ),
    clipBehavior: Clip.antiAlias,
    child: InkWell(
      onTap: () {
        Navigator.pushNamed(
          context,
          '/product-detail',
          arguments: product.id,
        );
      },
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          // Image Container
          Expanded(
            child: Container(
              width: double.infinity,
              color: isDark
                ? AppColors.darkBackground
                : AppColors.lightBackground,
              padding: EdgeInsets.all(8.r),
              // Image.network for caching
              child: Image.network(
                product.thumbnail,
                fit: BoxFit.contain,
                loadingBuilder: (context, child, loadingProgress) {
                  if (loadingProgress == null) return child;
                  return Center(
                    child: CircularProgressIndicator(
                      strokeWidth: 2,
                      value: loadingProgress.expectedTotalBytes != null
                        ? loadingProgress.cumulativeBytesLoaded /
                          loadingProgress.expectedTotalBytes!
                        : null,
                    ),
                  );
                },
              ),
          ),
          errorBuilder: (context, error, stackTrace) => Center(
            child: Icon(
              Icons.broken_image,
              color: isDark
                ? AppColors.darkMuted
                : AppColors.lightMuted,
            ),
          ),
        ],
      ),
    ),
  );

  // Text Info
  Padding(
    padding: EdgeInsets.all(10.r),
    child: Column(
      crossAxisAlignment: CrossAxisAlignment.start,
      children: [
        // Tag showing the brand name or product category
        Container(
          padding: EdgeInsets.symmetric(horizontal: 6.w, vertical: 2.h),
          decoration: BoxDecoration(
            color: theme.colorScheme.primaryContainer,
            borderRadius: BorderRadius.circular(4.r),
          ),
          child: CustomText(
            text: product.brand.isEmpty ? product.category.toUpperCase(),
            fontSize: 9.sp,
            fontWeight: FontWeight.bold,
            color: theme.colorScheme.onPrimaryContainer,
          ),
        ),
        SizedBox(height: 4.h),
        // Title
        SizedBox(
          height: 36.h,
          child: CustomText(
            text: product.title,
            fontSize: 13.sp,
            fontWeight: FontWeight.w600,
            maxLines: 2,
            overflow: TextOverflow.ellipsis,
          ),
        ),
        SizedBox(height: 4.h),
      ],
    ),
  );
}
```



```
// Show the star icon, average rating, and count of reviews
Row(
  children: [
    Icon(
      Icons.star_rounded,
      color: AppColors.ember,
      size: 14.sp,
    ),
    SizedBox(width: 2.w),
    CustomText(
      text: product.averageReviewRating.toStringAsFixed(1),
      fontSize: 11.sp,
      fontWeight: FontWeight.bold,
      color: isDark ? AppColors.sand : AppColors.emberDeep,
    ),
    SizedBox(width: 4.w),
    CustomText(
      text: '${product.reviews.length}',
      fontSize: 11.sp,
      color: isDark ? AppColors.darkMuted : AppColors.lightMuted,
    ),
  ],
),
SizedBox(height: 6.h),
// Price row
CustomText(
  text: '\${product.price.toStringAsFixed(2)}',
  fontSize: 14.sp,
  fontWeight: FontWeight.bold,
  color: theme.colorScheme.primary,
),
],
),
),
],
),
),
);
}

// Builds a skeleton loading screen
Widget _buildSkeletonGrid(bool isDark) {
  return GridView.builder(
    padding: EdgeInsets.symmetric(horizontal: 16.w, vertical: 8.h),
    gridDelegate: SliverGridDelegateWithFixedCrossAxisCount(
      crossAxisCount: 2,
      childAspectRatio: 0.65,
      crossAxisSpacing: 12.w,
      mainAxisSpacing: 12.h,
    ),
    itemCount: 6,
    itemBuilder: (context, index) {
      return Container(
        // Placeholder outer card shell
        decoration: BoxDecoration(
          color: isDark ? AppColors.darkCard : AppColors.lightCard,
          borderRadius: BorderRadius.circular(16.r),
          border: Border.all(
            color: isDark ? AppColors.darkBorder : AppColors.lightBorder,
          ),
        ),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            // Placeholder for the product image
            Expanded(
              child: Container(
                margin: EdgeInsets.all(12.r),
                decoration: BoxDecoration(
                  color: isDark ? AppColors.darkSkeleton : AppColors.lightSkeleton,
                  borderRadius: BorderRadius.circular(12.r),
                ),
              ),
            ),
            Padding(
              padding: EdgeInsets.all(12.r),
              child: Column(
                crossAxisAlignment: CrossAxisAlignment.start,
                children: [
                  // Placeholder for brand / category tag
                  Container(
                    width: 50.w,
                    height: 10.h,
                    color: isDark ? AppColors.darkBorder : AppColors.lightBorder,
                  ),
                  SizedBox(height: 8.h),
                  // Placeholder for product title (first line)
                  Container(
                    width: 120.w,
                    height: 12.h,
                    color: isDark ? AppColors.darkBorder : AppColors.lightBorder,
                  ),
                  SizedBox(height: 4.h),
                  // Placeholder for product title (second line)
                  Container(
                    width: 80.w,
                    height: 12.h,
                    color: isDark ? AppColors.darkBorder : AppColors.lightBorder,
                  ),
                  SizedBox(height: 12.h),
                  // Placeholder for product price
                  Container(
                    width: 60.w,
                    height: 14.h,
                    color: isDark ? AppColors.darkBorder : AppColors.lightBorder,
                  ),
                ],
              ),
            ),
          ],
        ),
      );
    },
  );
}
```



```
import package:flutter/material.dart;
import package:flutter/services.dart;
import package:flutter/cupertino.dart;
import package:flutter/widgets.dart;
import 'constants.dart';
import 'models/product.dart';
import 'providers/product_provider.dart';
import 'widgets/custom_text.dart';

class ProductDetailsScreen extends StatelessWidget {
  const ProductDetailsScreen({super.key});

  @override
  Widget build(BuildContext context) => ProductDetailsScreenSlate();

class ProductDetailsScreenSlate extends StatelessWidget {
  final ProductService _productService = ProductService();
  Product? _product;
  bool _isLoading = true;
  String _errorMessage = '';
  bool _isFavorite = false;

  bool _isFavorite = false;

  void didChangeDependencies() {}
  void initState() {
    // 1. Initial state
    final productId = ModalRoute.of(context)?.settings.arguments as int;
    loadProductDetails(productId);
    final productId = ModalRoute.of(context)?.settings.arguments as int;
    loadProductDetails(productId);
  }

  Future<void> _loadProductDetails(int id) async {
    setState(() {
      _isLoading = true;
      _errorMessage = '';
    });
    try {
      final product = await _productService.getProductById(id);
      setState(() {
        _product = product;
        _isLoading = false;
      });
    } catch (e) {
      setState(() {
        _isLoading = false;
        _errorMessage = e.toString();
      });
    }
  }

  @override
  Widget build(BuildContext context) {
    // Get color theme information from context
    final theme = Theme.of(context);
    final isDark = Theme.brightness == Brightness.dark;

    if (_isLoading) {
      return Scaffold(
        backgroundColor: theme.scaffoldBackgroundColor,
        appBar: AppBar(
          backgroundColor: Colors.transparent,
          elevation: 0,
          title: Text(
            'Loading...',
            style: TextStyle(
              color: isDark ? AppColors.sand : AppColors.tealDess,
              opacity: 0.5,
            ),
          ),
        ),
      );
    }

    if (_errorMessage.isNotEmpty || _product == null) {
      return Scaffold(
        backgroundColor: theme.scaffoldBackgroundColor,
        appBar: AppBar(
          backgroundColor: Colors.transparent,
          elevation: 0,
          title: Text(
            'Error: ${_errorMessage}',
            style: TextStyle(
              color: isDark ? AppColors.sand : AppColors.tealDess,
              opacity: 0.5,
            ),
          ),
        ),
      );
    }

    final product = _product;

    return Scaffold(
      backgroundColor: theme.scaffoldBackgroundColor,
      body: CustomScrollView(
        slivers: [
          // Elegant transparent App Bar with Hero Image
          SliverAppBar(
            expandedHeight: 200.h,
            pinned: true,
            stretch: true,
            leading: Padding(
              padding: EdgeInsets.all(8.r),
              child: CircleAvatar(
                backgroundImage: NetworkImage(
                  product?.image ?? '',
                ),
                radius: 30.r,
                backgroundColor: Colors.transparent,
                elevation: 0,
              ),
            ),
            title: Text(
              product?.name ?? '',
              style: TextStyle(
                color: isDark ? AppColors.sand : AppColors.tealDess,
                opacity: 0.5,
              ),
            ),
            actions: [
              Padding(
                padding: EdgeInsets.all(8.r),
                child: CircleAvatar(
                  radius: 30.r,
                  backgroundColor: isDark
                    ? AppColors.darKCard.v1(Values.alpha: 0.75)
                    : AppColors.lightCard.v1(Values.alpha: 0.8),
                  child: Text(
                    product?.isFavorite ?? '0',
                    style: TextStyle(
                      color: isDark ? AppColors.sand : AppColors.tealDess,
                      opacity: 0.5,
                    ),
                  ),
                ),
              ),
            ],
          ),
          SliverList(
            children: [
              // Product Details
              Padding(
                padding: EdgeInsets.all(8.r),
                child: Column(
                  crossAxisAlignment: CrossAxisAlignment.start,
                  children: [
                    // Product Name
                    Text(
                      product?.name ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Price
                    Text(
                      product?.price ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Description
                    Text(
                      product?.description ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Image
                    Image(
                      image: NetworkImage(
                        product?.image ?? '',
                      ),
                      width: 100.w,
                      height: 100.h,
                      fit: BoxFit.cover,
                    ),
                    // Product Status
                    Text(
                      product?.status ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Category
                    Text(
                      product?.category ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Tags
                    Text(
                      product?.tags ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Reviews
                    Text(
                      product?.reviews ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Comments
                    Text(
                      product?.comments ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Likes
                    Text(
                      product?.likes ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Shares
                    Text(
                      product?.shares ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Favorites
                    Text(
                      product?.favorites ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product History
                    Text(
                      product?.history ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Settings
                    Text(
                      product?.settings ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Notifications
                    Text(
                      product?.notifications ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Alerts
                    Text(
                      product?.alerts ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Announcements
                    Text(
                      product?.announcements ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Updates
                    Text(
                      product?.updates ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Changes
                    Text(
                      product?.changes ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Actions
                    Text(
                      product?.actions ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Interactions
                    Text(
                      product?.interactions ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Engagement
                    Text(
                      product?.engagement ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Metrics
                    Text(
                      product?.metrics ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Analytics
                    Text(
                      product?.analytics ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Insights
                    Text(
                      product?.insights ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Trends
                    Text(
                      product?.trends ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Forecasts
                    Text(
                      product?.forecasts ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Projections
                    Text(
                      product?.projections ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Scenarios
                    Text(
                      product?.scenarios ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Simulations
                    Text(
                      product?.simulations ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Models
                    Text(
                      product?.models ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Hypotheses
                    Text(
                      product?.hypotheses ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Theories
                    Text(
                      product?.theories ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Concepts
                    Text(
                      product?.concepts ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Principles
                    Text(
                      product?.principles ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Rules
                    Text(
                      product?.rules ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Guidelines
                    Text(
                      product?.guidelines ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Standards
                    Text(
                      product?.standards ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Practices
                    Text(
                      product?.practices ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Procedures
                    Text(
                      product?.procedures ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Processes
                    Text(
                      product?.processes ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Methods
                    Text(
                      product?.methods ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Techniques
                    Text(
                      product?.techniques ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Strategies
                    Text(
                      product?.strategies ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Tactics
                    Text(
                      product?.tactics ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Approaches
                    Text(
                      product?.approaches ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Frameworks
                    Text(
                      product?.frameworks ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Systems
                    Text(
                      product?.systems ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Structures
                    Text(
                      product?.structures ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Frameworks
                    Text(
                      product?.frameworks ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Systems
                    Text(
                      product?.systems ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Structures
                    Text(
                      product?.structures ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Frameworks
                    Text(
                      product?.frameworks ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Systems
                    Text(
                      product?.systems ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 0.5,
                      ),
                    ),
                    // Product Structures
                    Text(
                      product?.structures ?? '',
                      style: TextStyle(
                        color: isDark ? AppColors.sand : AppColors.tealDess,
                        opacity: 
```

[illegible]



```
Widget _buildSpecRow(String label, String value) {
  return Padding(
    padding: EdgeInsets.symmetric(vertical: 4.h),
    child: Row(
      crossAxisAlignment: CrossAxisAlignment.start,
      children: [
        SizedBox(
          width: 110.w,
          child: CustomText(
            text: label,
            fontSize: 13.sp,
            fontWeight: FontWeight.w500,
          ),
        ),
        Expanded(
          child: CustomText(
            text: value,
            fontSize: 13.sp,
          ),
        ),
      ],
    ),
  );
}

String _formatDate(String dateStr) {
  try {
    final dateTime = DateTime.parse(dateStr);
    return '${dateTime.year}-${dateTime.month.toString().padLeft(2, '0')}-${dateTime.day.toString().padLeft(2, '0')}';
  } catch (_) {
    return dateStr;
  }
}

// skeleton loading
Widget _buildSkeletonDetail(ThemeData theme, bool isDark) {
  return SingleChildScrollView(
    child: Column(
      crossAxisAlignment: CrossAxisAlignment.start,
      children: [
        // Placeholder for the product hero image
        Container(
          height: 350.h,
          width: double.infinity,
          color: theme.cardTheme.color,
          child: Center(
            child: Icon(
              Icons.image,
              size: 80.sp,
              color: isDark ? AppColors.darkSkeleton : AppColors.lightSkeleton,
            ),
          ),
        ),
        Padding(
          padding: EdgeInsets.all(16.r),
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              // Placeholder for category / brand tag
              Container(
                width: 100.w,
                height: 20.h,
                decoration: BoxDecoration(
                  color: isDark ? AppColors.darkSkeleton : AppColors.lightSkeleton,
                  borderRadius: BorderRadius.circular(4.r),
                ),
              ),
              SizedBox(height: 12.h),
              // Placeholder for product title
              Container(
                width: 250.w,
                height: 24.h,
                decoration: BoxDecoration(
                  color: isDark ? AppColors.darkSkeleton : AppColors.lightSkeleton,
                  borderRadius: BorderRadius.circular(4.r),
                ),
              ),
              SizedBox(height: 8.h),
              // Placeholder for product price
              Container(
                width: 80.w,
                height: 24.h,
                decoration: BoxDecoration(
                  color: isDark ? AppColors.darkSkeleton : AppColors.lightSkeleton,
                  borderRadius: BorderRadius.circular(4.r),
                ),
              ),
              SizedBox(height: 12.h),
              // Placeholder for stock status indicator
              Row(
                children: [
                  Container(
                    width: 18.w,
                    height: 18.h,
                    decoration: BoxDecoration(
                      color: isDark ? AppColors.darkSkeleton : AppColors.lightSkeleton,
                      shape: BoxShape.circle,
                    ),
                  ),
                  SizedBox(width: 4.w),
                  Container(
                    width: 120.w,
                    height: 14.h,
                    decoration: BoxDecoration(
                      color: isDark ? AppColors.darkSkeleton : AppColors.lightSkeleton,
                      borderRadius: BorderRadius.circular(4.r),
                    ),
                  ),
                ],
              ),
              Divider(
                height: 32.h,
                thickness: 1,
                color: isDark ? AppColors.darkBorder : AppColors.lightBorder,
              ),
              // Placeholder for description header
              Container(
                width: 100.w,
                height: 16.h,
                decoration: BoxDecoration(
                  color: isDark ? AppColors.darkSkeleton : AppColors.lightSkeleton,
                  borderRadius: BorderRadius.circular(4.r),
                ),
              ),
              SizedBox(height: 8.h),
              // Placeholder for description text lines
              Column(
                children: List.generate(
                  3,
                  (index) => Padding(
                    padding: EdgeInsets.symmetric(vertical: 4.h),
                    child: Container(
                      width: double.infinity,
                      height: 14.h,
                      decoration: BoxDecoration(
                        color: isDark ? AppColors.darkSkeleton : AppColors.lightSkeleton,
                        borderRadius: BorderRadius.circular(4.r),
                      ),
                    ),
                  ),
                ),
              ),
              SizedBox(height: 24.h),
              // Placeholder for specification details card
              Container(
                width: double.infinity,
                height: 120.h,
                decoration: BoxDecoration(
                  color: theme.cardTheme.color,
                  borderRadius: BorderRadius.circular(12.r),
                  border: Border.all(
                    color: isDark ? AppColors.darkBorder : AppColors.lightBorder,
                  ),
                ),
              ),
            ],
          ),
        ),
      ],
    ),
  );
}
```

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

import '../poviders/theme_provider.dart';

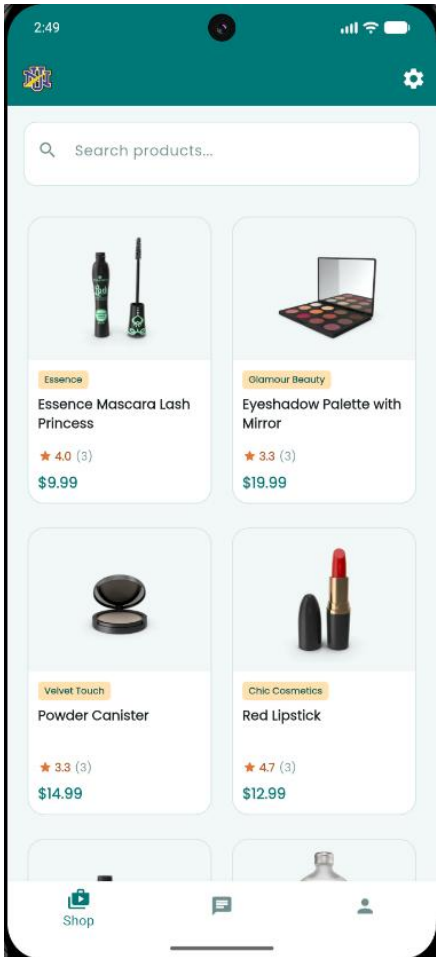
class SettingsScreen extends StatelessWidget {
  const SettingsScreen({super.key});

  @override
  Widget build(BuildContext context) {
    final themeProvider = context.watch<ThemeProvider>();

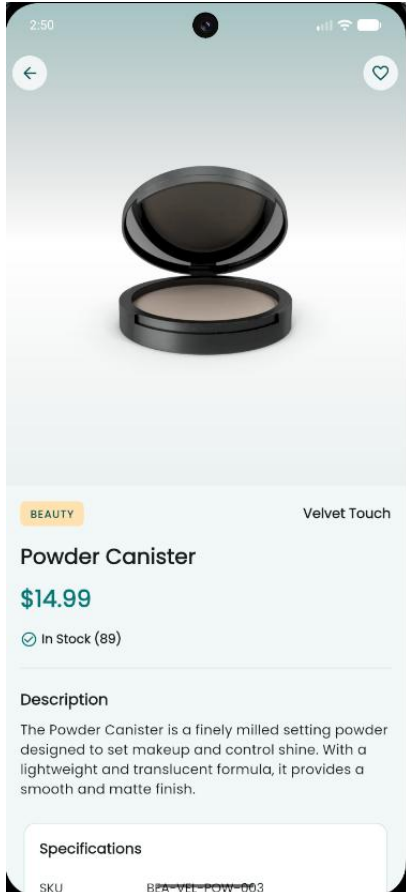
    return Scaffold(
      appBar: AppBar(title: const Text('Theme Settings')),
      body: SwitchListTile(
        title: const Text('Dark Mode'),
        value: themeProvider.isDark,
        onChanged: (_) => themeProvider.toggleTheme(),
      ),
    );
  }
}
```

Output – [name of the output]

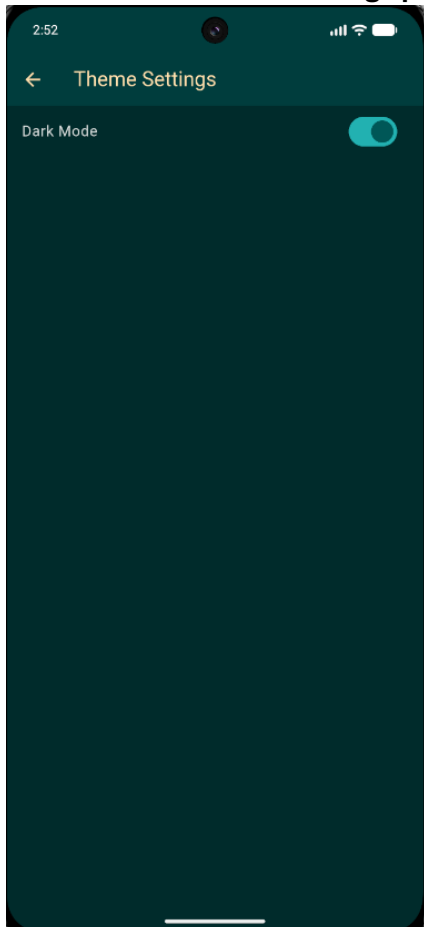
Enhancement 1: Add search bar above the article list.



Enhancement 2: Add details page when clicked the card.



Enhancement 3: Add settings page to move the dark/light mode switch.



Discussion

The Product model is used to organize the data from the API. It uses `Product.fromJson()` to convert the JSON data into a Dart object. The `ProductService` is responsible for getting the data from the API using `http.get()` and returning a `List<Product>`, so the UI doesn't need to handle network requests. The `ProductScreen` calls the service in `initState()`, then uses `setState()` to update the screen and display the products.

The design pattern used is separation of concerns, where each file has its own responsibility. This makes the code easier to understand and maintain. It also uses a factory constructor for JSON conversion, `Future` and `async/await` for loading data without freezing the app, and the `Provider` pattern to manage the app's theme.

Lab Activity Rubric {*Copy and paste the Rubric here*}

Name	Jamaine Grace Tuazon
Section	INF231
Date	August 04, 2026

Criteria	Excellent (4 pts)	Good (3 pts)	Fair (2 pts)	Poor (1 pt)	Points
Code Implementation	Code runs flawlessly, includes all required features (placeholders, avatars, widgets for like/comment).	Code mostly correct, minor errors, or missing one minor feature.	Noticeable errors or missing two or more features.	Code does not run or is significantly incomplete.	
Design Consistency	UI closely resembles intended layout, proper spacing, alignment, and styling.	Layout mostly accurate, minor spacing or alignment issues.	Layout deviates significantly from the intended design or appears incomplete.	Layout is unrecognizable or does not follow the instructions.	
Use of Widgets and Enhancements	Properly implements widgets for placeholders, avatars, and like/comment buttons.	Most widgets are correctly implemented, with minor issues or missing one enhancement.	Widgets are inconsistently used or missing two or more enhancements.	Widgets are not used, or enhancements are not implemented.	
Code Organization and Readability	Code is well-structured, follows best practices, and adheres to the specified file organization.	Code is mostly organized and readable, minor deviations from the required structure.	Code is poorly organized or does not follow the specified structure.	Code is disorganized and difficult to understand.	
Back-End Integration	Back-end integration works flawlessly with efficient API calls, secure data handling, and real-time updates; no errors or downtime.	Back-end integration is mostly correct with only minor API, data, or response handling issues.	Back-end integration has noticeable flaws or inefficiencies; multiple API calls fail or data handling is inconsistent.	Back-end integration has major issues; frequent failures, poor response handling, or insecure practices.	
Other Comments/ Observations:				Total Score	
				Rating: (Total Score/20) * 100	